

Heredity

NCERT Class 10 Science Chapter 8 (Hindi Detailed Notes)
NCERT Class 10 Science Ch 8 | English Detailed Notes

COLOUR CODE: RED = Core Basics / extra help (Class 7-8-9 wali cheezein jo yaad honi chahiye).

COLOUR CODE: GREEN = Board exam me ye topics BAAR-BAAR aate hain - pakka yaad karo.

Baaki normal black text = main chapter content (detail me, simple bhasha me).

CORE BASICS - PEHLE YE PADHO (CLASS 7-8-9 KA FOUNDATION)

Tune 7-8-9 nahi padhi - koi tension nahi. Ye page is chapter ki base hai.
Ek baar achhe se padh le, fir poora chapter aasaani se samajh aayega.

1) CELL aur NUCLEUS (Class 8 ka idea):

- Cell = jeev ki sabse chhoti unit (body ki "eent"/brick).
- Nucleus = cell ka "control room" (beech me). Isi me sari genetic information (DNA) rehti hai jo trait decide karti hai.

2) DNA, CHROMOSOME, GENE (sabse zaroori - intro):

- DNA = ek lamba thread-jaisa molecule jisme "instructions" likhe hote.
- CHROMOSOME = DNA jab tightly lipti/packed hoti to thread-jaisi structure banti = chromosome. Nucleus me jodi (pairs) me hote hain.
- GENE = DNA/chromosome ka ek chhota tukda jo EK trait control karta (jaise height ka gene, seed colour ka gene).

3) GAMETE (sex cell):

- Reproduction ke liye banne wali special cell = gamete.
- Male gamete = sperm (shukraanu), Female gamete = egg/ovum (andaanu).
- Gamete me normal cell se AADHE (half) chromosome hote hain.

4) FERTILISATION aur ZYGOTE:

- Sperm + egg ka milna = FERTILISATION.
- Milne ke baad bani pehli single cell = ZYGOTE. Yahi badh ke baby banti.
- Zygote me dono parents ke chromosome aa jaate (aadhe maa + aadhe baap se).

5) TRAIT / CHARACTER kya hai:

- Trait = jeev ka koi feature/lakshan (height, eye colour, seed shape).
- Ye traits genes ke through parents se bacchon me aate hain.

6) DOMINANT vs RECESSIVE (simple intro):

- DOMINANT = jo trait dikh jaata hai (express ho jaata), bhale doosra bhi ho.
- RECESSIVE = jo dab jaata hai, dikhta tabhi jab dono allele recessive ho.

7) VARIATION:

- Ek hi species ke jeevon me chhote-chhote farak (differences) = variation.
- Reproduction (khaas kar sexual) se variation aata hai - sab bilkul same nahi hote.

8) SEX CHROMOSOMES (XX / XY) ka idea:

- Kuch chromosome jeev ka GENDER decide karte = sex chromosomes.
- Insaan me: female = XX, male = XY. (Detail aage main content me.)

9) MITOSIS vs MEIOSIS (simple idea):

- MITOSIS = normal body cell division; chromosome number SAME rehta (growth/repair me kaam aata).
- MEIOSIS = gamete banane wali division; chromosome number AADHA ho jaata (isiliye gamete me half chromosome hote hain).

1. HEREDITY aur INHERITED TRAITS

HEREDITY = parents se bacchon me traits (lakshan) ka transfer hona.

- Inheritance/Anuvanshikta wahi process hai.

Bachhe apne parents jaise kyun dikhte hain?

- Kyunki parents apne GENES (DNA) gametes (sperm/egg) ke through bacche ko dete hain. Zygote me dono parents ke gene aate, isliye bacche me dono ki jhalak dikhti.
- Har trait ke liye bacche ko 2 copy (allele) milte: 1 maa se, 1 baap se.

INHERITED TRAIT = jo gene ke through mile (height, eye colour, blood group).

Ye hi traits aage ki generation me jaate hain.

2. MENDEL aur unke MATAR (PEA PLANT) EXPERIMENTS

Gregor Mendel ko "Father of Genetics" kehte hain. Unhone garden pea plant (scientific naam: *Pisum sativum*) par cross karke inheritance ke rules diye.

2.1 MENDEL NE PEA PLANT HI KYUN CHUNA

Ye exam me favourite question hai - reasons yaad rakho:

- (i) Pea me CLEAR contrasting traits the (tall/short, round/wrinkled seed, green/yellow seed) - aasaani se pehchaan aate.
- (ii) Cross karna (cross-pollination) aasaan + self-pollination bhi ho sakti, to Mendel control kar sakte the.
- (iii) Life cycle CHHOTA tha aur ek baar me bahut saare seeds milte the (zyada data + jaldi results).

2.2 KUCH ZAROORI TERMS (DEFINITIONS)

GENE = DNA ka tukda jo ek trait control karta.

ALLELE = ek hi gene ke alag-alag roop (jaise T = tall allele, t = short allele).

GENOTYPE = jeev ka genetic make-up (allele ka combination), jaise TT, Tt, tt.

PHENOTYPE = jo bahar DIKHTA hai (physical roop), jaise "tall" ya "short".

DOMINANT = jo allele express ho jaata (capital letter, e.g. T).

RECESSIVE = jo dab jaata, sirf double recessive me dikhta (small letter, e.g. t).

HOMOZYGOUS = dono allele SAME (TT ya tt) - ise "pure" bhi kehte.

HETEROZYGOUS = dono allele ALAG (Tt) - ise "hybrid" bhi kehte.

F1 GENERATION = parents (P) ke cross se bani pehli santaan (first filial).

F2 GENERATION = F1 ko aapas me cross/self karne se bani agli santaan.

3. MONOHYBRID CROSS (EK TRAIT)

MONOHYBRID = ek hi trait (jaise height) ko dekhte hue cross. Bahut important.

Cross: pure Tall (TT) x pure short (tt)

- Tall ke gamete = T, short ke gamete = t.
- F1 me sab milte = Tt -> SAARE TALL (kyunki T dominant, t recessive).
- Matlab F1 me short trait GAYAB ho gaya (par gene chhupa hua hai).

Ab F1 ko self/cross karo: Tt x Tt

- Gametes: T aur t (dono parents se).

PUNNETT SQUARE (F1 x F1):

	T	t
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T	TT	Tt
---	----	----

t	Tt	tt
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F2 RESULTS (RATTA maaro):

- PHENOTYPE ratio = 3 Tall : 1 short = 3:1
- GENOTYPE ratio = 1 TT : 2 Tt : 1 tt = 1:2:1
- Short trait (tt) F2 me WAPAS aa gaya -> saabit hua gene chhupa tha (recessive).

Note: 3 tall me se 1 pure (TT) aur 2 hybrid (Tt) hote hain.

4. DIHYBRID CROSS (DO TRAITS)

DIHYBRID = ek saath DO traits dekhte hue cross. Ratio 9:3:3:1 important hai.

Cross: Round-Yellow seed (RRYY) x wrinkled-green seed (rryy)
- F1 sab = RrYy -> sab Round aur Yellow (R aur Y dominant hain).

Ab F1 self karo: RrYy x RrYy
- Har parent ke 4 type ke gamete: RY, Ry, rY, ry.

PUNNETT SQUARE (16 boxes):

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F2 PHENOTYPE ratio = 9 : 3 : 3 : 1

- 9 Round-Yellow : 3 Round-green : 3 wrinkled-Yellow : 1 wrinkled-green.

CONCLUSION: do traits ALAG-ALAG (INDEPENDENTLY) inherit hote hain - ek trait doosre ko affect nahi karta. Isi se naye combos (Round-green, wrinkled-Yellow) bante jo parents me alag the.

5. INHERITANCE KE RULES aur TRAIT KAISE EXPRESS HOTA

5.1 RULES OF INHERITANCE (Mendel se)

- Traits genes ke through control hote hain.
- Har trait ke 2 allele hote; har PARENT ek allele deta (gamete ke through).
- Heterozygous (Tt) me DOMINANT allele express hota, recessive dab jaata.
- Gamete banne par allele alag ho jaate (ek hi allele gamete me jaata).

5.2 TRAIT KAISE EXPRESS HOTA HAI (GENE -> PROTEIN -> TRAIT)

- Ek gene asal me ek PROTEIN (ya enzyme) banane ka code rakhta hai.
 - Wahi protein/enzyme body me kaam karke trait banata hai.
- Example (simple): plant me "tallness" ka ek gene enzyme banata jo hormone zyada banata -> plant tall. Agar allele aisa ho jo enzyme kam-asar banaye to plant short reh jaata.

- Short me: GENE -> PROTEIN/ENZYME -> TRAIT. Dominant allele wala efficient protein banata, isliye uska trait dikh jaata.

6. SEX DETERMINATION IN HUMANS (LADKA YA LADKI)

Insaan me 23 PAIR (yaani 46) chromosomes hote hain.

- In me se 22 pair = autosomes (body wale).
- 1 pair = SEX CHROMOSOMES (gender decide karte).

FEMALE = XX , MALE = XY.

Gametes me kya jaata hai?

- Maa (XX) ke saare eggs me sirf X aata (mother HAMESHA X deti).
- Baap (XY) ke sperm me ya to X ya Y aata.

PUNNETT SQUARE (Mother XX x Father XY):

	X	X
X	XX	XX
Y	XY	XY

RESULT: agar baap ka X wala sperm jeeta -> XX -> LADKI.

agar baap ka Y wala sperm jeeta -> XY -> LADKA.

Matlab BACCHE KA GENDER BAAP KE SPERM se decide hota (50-50 chance).

Isiliye ladki ke liye maa ko blame karna galat aur unscientific hai.

7. ACQUIRED vs INHERITED TRAITS

Ye difference exam me bahut poocha jaata - dono ko alag samjho.

INHERITED (genetic) TRAITS:

- Gene/DNA ke through parents se mile (eye colour, blood group, height).
- Ye AGLI generation me jaate hain (kyunki gametes/DNA me hote hain).

ACQUIRED TRAITS:

- Jeevan ke dauraan body par aaye changes (kam weight starvation se, kati hui tail/dumka, body building se bani muscles, seekhi hui language).
- Ye AGLI generation me NAHI jaate.

KYUN NAHI JAATE (reason - yaad rakho):

- Acquired changes sirf BODY (somatic) cells ko affect karte hain, ye germ/reproductive cells ke DNA ko change NAHI karte.
- Gamete ka DNA hi agli pidhi me jaata - jab wahi nahi badla, to acquired trait pass nahi hota.

Classic example: agar chuhe (mouse) ki tail kaat do, to bhi uske bacche poori tail ke saath hi paida honge (kaatna DNA nahi badalta).

8. SOLVED EXAMPLES (HARDEST -> EASIEST)

Ye type ke questions exam me pakka aate - steps khud likhke practice karo.

EXAMPLE 1 (Hardest): Dihybrid cross karke dikhao ki F2 me 9:3:3:1 kaise aata aur isse kya conclusion nikalta?

- Parents: RRYy (round-yellow) x rryy (wrinkled-green).
- F1: sab RrYy (round-yellow) kyunki R aur Y dominant.
- F1 x F1 -> 4 gametes (RY, Ry, rY, ry) -> 16 box wala Punnett square.
- F2 phenotype: 9 round-yellow, 3 round-green, 3 wrinkled-yellow, 1 wrinkled-green = 9:3:3:1.
- Conclusion: do traits INDEPENDENTLY inherit hote (ek doosre ko affect nahi).

EXAMPLE 2: Tall (TT) x short (tt) ka monohybrid cross Punnett square se dikhao; F2 ka phenotype aur genotype ratio batao.

- F1: sab Tt (tall). F1 self -> Tt x Tt.

	T	t
T	TT	Tt
t	Tt	tt

- Phenotype = 3 tall : 1 short = 3:1.
- Genotype = 1 TT : 2 Tt : 1 tt = 1:2:1.

EXAMPLE 3: Cross Tt x tt diya hai. Offspring ka genotype aur phenotype ratio?

	T	t
t	Tt	tt
t	Tt	tt

- Genotype = 2 Tt : 2 tt = 1:1. Phenotype = 1 tall : 1 short = 1:1.

EXAMPLE 4: Cross dikha ke explain karo bacche ka sex (gender) baap se kaise tay hota hai.

- Mother XX (sirf X deti), Father XY (X ya Y deta).

	X	X
X	XX	XX
Y	XY	XY

- XX = ladki, XY = ladka. Y baap ke sperm se aata, isliye gender BAAP decide karta (50-50 chance).

EXAMPLE 5: Tall x short cross me F1 sab tall the, aur F2 me 3 tall : 1 short mile. Konsa trait dominant aur konsa recessive? Kaise pata chala?

- Dominant = TALL (F1 me sirf yahi dikha).
- Recessive = SHORT (F1 me chhup gaya, F2 me wapas 1/4 me aaya).
- Jo F1 me dikhe woh dominant; jo chhup ke F2 me laute woh recessive.

EXAMPLE 6: Ek aadmi ne body-building se strong muscles banaye. Kya ye trait uske bacchon me jaayega? Kyun?

- NAHI jaayega. Ye ACQUIRED trait hai - sirf body (somatic) cells par asar,

germ cells/gamete ke DNA me koi change nahi. Isliye inherit nahi hota.

EXAMPLE 7 (Easiest): Mendel ne apne experiment ke liye pea plant hi kyun chuna?

- Clear contrasting traits (tall/short, round/wrinkled), cross/self dono aasaan, aur chhota life cycle + bahut saare seeds (jaldi + zyada data).

9. QUICK Q&A - PADHNE KE BAAD KHUD CHECK KARO

Pehle khud answer socho, fir niche milao. Ye board-style questions hain.

Q1. Heredity kise kehte hain?

A1. Parents se santaan me traits (lakshan) ka genes ke through transfer hona.

Q2. Genotype aur phenotype me farak?

A2. Genotype = andar ka genetic make-up (TT/Tt/tt). Phenotype = bahar dikhne wala roop (tall/short).

Q3. Homozygous aur heterozygous me farak?

A3. Homozygous = dono allele same (TT ya tt). Heterozygous = dono allele alag (Tt).

Q4. Monohybrid cross ka F2 phenotype aur genotype ratio?

A4. Phenotype 3:1 (tall:short); genotype 1:2:1 (TT:Tt:tt).

Q5. Dihybrid cross ka F2 phenotype ratio aur conclusion?

A5. 9:3:3:1; conclusion - do traits independently inherit hote hain.

Q6. Insaan me ladki paida hone ke liye baap ne konsa chromosome diya?

A6. X chromosome (X + X = XX = ladki). Y se ladka banta.

Q7. Bacche ka gender kaun decide karta hai aur kyun?

A7. Baap (father), kyunki maa hamesha X deti par baap X ya Y - jo sperm jeete wahi gender tay karta.

Q8. Acquired trait kya hai? Ek example.

A8. Jeevan me body par aaya change (jaise kati hui tail / banayi hui muscles).
Ye agli generation me nahi jaata.

Q9. Acquired traits agli pidhi me kyun nahi jaate?

A9. Kyunki ye germ/reproductive cells ke DNA ko change nahi karte; sirf gamete ka DNA aage jaata hai.

Q10. Dominant aur recessive allele me farak?

A10. Dominant express ho jaata hai (Tt me bhi dikhta); recessive tabhi dikhta jab dono allele recessive hon (tt).

EXAM STRATEGY (1 minute revision):

- Monohybrid 3:1 (phenotype) + 1:2:1 (genotype) Punnett square se aana chahiye.
 - Dihybrid 9:3:3:1 + "independent inheritance" conclusion.
 - Genotype/phenotype, dominant/recessive, homozygous/heterozygous definitions.
 - Sex determination XX/XY - baap gender decide karta (cross banana aana chahiye).
 - Acquired vs inherited traits + reason (germ cell DNA nahi badalta).
 - Mendel ne pea kyun chuna - 3 reasons.
 - CORE BASICS page (gene, DNA, gamete, zygote) bhool jaao to wapas pehle padho
 - base strong rakho.
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